

Dream Garden: Exploring Location-Based, Collaboratively-Created Augmented Reality Spaces

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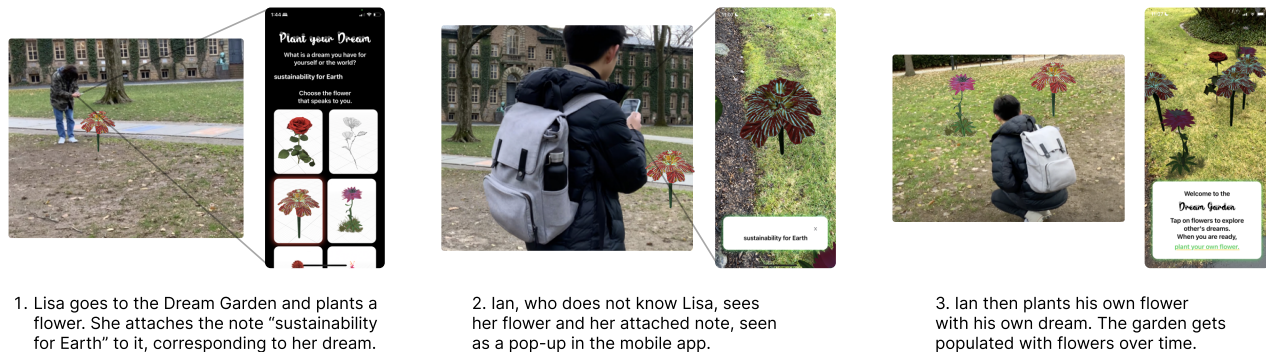


Figure 1: Description of the Dream Garden system in use.

ABSTRACT

We introduce Dream Garden, an augmented reality application (AR) that lets people place 3D flowers into the physical world to build a collaborative location-based garden. Despite the potential for connecting strangers in the digital realm, current research has not explored location-based augmented reality experiences that enable strangers to connect by building artifacts collaboratively. We explore this by creating an AR digital community garden deployed in a specific location. Anyone with the proper app can access and see the flowers previously planted by strangers, as well as plant their own flowers to grow the garden. We evaluated this app with 10 participants, with 5 visiting the digital garden more than once, to evaluate their sense of connection to the other participants as each participant added one digital flower to the garden. We found that participants were joyful about building a shared space, were excited about the dynamic nature of the garden, felt a connection to the physical location of the digital garden, and expressed a sense of belonging to a community of strangers but not necessarily an emotional connection.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing devices.*

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KEYWORDS

location-based, augmented reality, collaboration, loneliness, connection, strangers

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1 INTRODUCTION

In this paper, we design and discuss Dream Garden, an application that uses a participatory, augmented reality (AR), and location-based experience to create a digital forum for strangers to interact and connect. This research is motivated by two mental well-being needs: the need for shared spaces [27], and the need for connecting with strangers [10]. Both needs are particularly timely because of an increase in loneliness in our society, i.e., around 60% of US adults today [12, 18]. Casually connecting with strangers allows us to feel like we are part of a community, and is vital for our feelings of social well-being [13]. Shared spaces contribute to social connectedness, which in turn, supports mental well-being [21, 25]. Individuals feel a sense of connectedness to others through (1) Sense of shared reality with another [14] (2) Experiencing positive emotions together [11] (3) Sharing of personal information with others [28] (4) Having shared goals [32]. People also develop strong emotional connections to certain locations, leading people to feel more connected and attuned to changes while in that location [15, 24].

Augmented reality allows us to create new shared spaces that connect strangers, unique in that AR extends the physical realm and

thus touches upon the ways people connect to each other in ways non-AR alternatives can't, such as creating a more direct sense of embodiment in a location and a shared digital reality. Motivated by the well-being need of connection and with the above research in mind, we created Dream Garden - a shared digital garden only available in a specific location that allows strangers to anonymously plant flowers that have an associated textual note of the stranger's dream into the garden for everyone else to see, having the digital garden be built up anonymously by strangers and encouraging them to share personal information.

We evaluated how Dream Garden (1) makes participants feel connected to each other, (2) their experience growing the garden together, (3) how they react to the dynamic nature of the garden, and (4) how the garden changed their connection to the physical location.

We learned that participants were excited by the dynamic nature of the space, and felt a sense of connection to the physical location of the garden and each other, but felt connected in different ways - some more emotionally, others through a sense of joy. Furthermore, we found that participants were invested in co-creating the garden with and expressed a desire to meet with other participants, showing that these types of AR experiences may act as initiators for in-person interaction.

2 RELATED WORK

People are increasingly disconnected, and alone [12, 31]. There are fewer opportunities to connect with strangers face-to-face [27]. Unfortunately, digital tech might contribute to isolation [27, 33]. Could we design tech differently? Despite its isolating features, digital tech can also help connect strangers face-to-face. For example, early LBS such as Foursquare, Gowalla, Meetup.com, and more recent ones like Pokemon Go bring people together in certain locations [1–3, 6]. However, these technologies do not allow strangers to collaborate in creating spaces together, which bonds strangers together through a shared reality and goal, which is what we set out to address.

Previous research has looked into location-based physical interventions, particularly called digital urban interventions, that aim to encourage strangers to connect through a shared goal. Jokebox, a system that encourages two strangers to coordinate with each other to hear a joke [9], and Pop-spot, a project that encourages strangers to come together to discuss their local transportation system [26], are two examples. Though these projects encourage strangers to interact through a shared goal, strangers do not contribute to shaping their direct environments, and their interactions with these systems are ephemeral.

Non-location-based digital examples of collaboratively built digital spaces include projects such as Breonna's Garden, an AR garden experience that cultivates a safe space online to process grief around Breonna Taylor's death and allows strangers to record messages of hope for Breonna's family [8], and NICE, a collaborative learning environment where kids build together [29]. Further, collaborative creation has been explored in the medium of virtual reality through social VR applications such as VRChat [16]. However, these projects are not location-based and thus do not encourage face-to-face interaction.

Architects and urban designers have theorized about physical spaces that are ever dynamic and change based on the input of strangers, such as Cedric Price's Fun Palace and Archigram's Plug-In-City [22, 23]. These ideas include structures that allow people to change their physical spaces to match their needs by moving walls, modules, ramps, and other building elements. Over time strangers start to collaborate to change their environment, creating an opportunity for people to bond over shared goals, such as creating spaces for performance. In the physical world, there are some basic examples of collaboratively built experiences that are built up through the interaction of strangers. Candy Chang's work and the Stranger's Project are examples of participatory shared spaces where strangers come together to create a shared space by writing their own stories and putting them on view for all to see, but they can't actively interact with each other [4, 5]. As they are physical, they are tied to specific locations where people congregate, such as Washington Sq Park in NYC. We try to implement these ideas of ever-changing, stranger-mediated spaces digitally through AR.

Location-based AR research has explored placing archival history in location, finding that this increased place attachment [20], and placing social media photos in location through AR has been found to have a similar effect [19]. However, these experiences operate across multiple locations and though users can add virtual content to locations, the shared goal of co-creating a single location and having strangers interact is not the focus.

Previous work in AR to address individuals building up spaces together that are location-based have been explored in projects such as Snap Local Lenses, MarbleVerse, and Blocks [7, 17, 30]. However, they do not explicitly focus on strangers collaborating, and instead are aimed at existing friend relationships. We focus on making Dream Garden a digital, location-based, collaboratively-built space that changes based on every strangers' interaction.

3 DREAM GARDEN SYSTEM

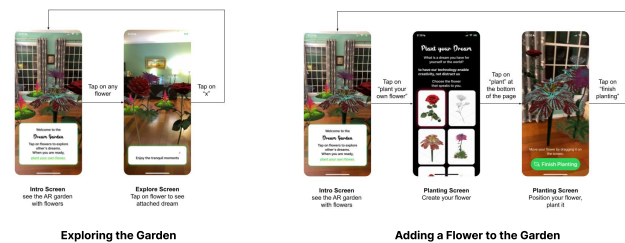


Figure 2: Interaction flows in the Dream Garden app, showing screenshots. Note screenshots are not taken in the exact location of the research.

Dream Garden enables a participant using a mobile phone in a certain geo-location to see a digital augmented reality garden made up of multiple flowers, superimposed onto the ground. The theme of the garden is “dreams,” and each flower is connected to a note that corresponds to an anonymous stranger’s dream. Participants can walk through the garden, and when they tap on a flower, the associated note appears on the phone display.

The theme of "dreams" and of a garden is arbitrary but chosen to unite strangers under a shared context everyone can understand and to encourage the sharing of personal information. A main consideration in the design of the system is that the digital AR space permanently changes every time a participant interacts with it. Here, the building blocks of the shared AR space are flowers, and every time a new participant visits the garden, they add a new flower that will persist. We chose a location of heavy foot traffic for the garden due to wanting the garden to be a place where people overlap and congregate.

A participant can "plant" a flower into the garden by entering a note corresponding to their own dream and then choosing one of seven types of flowers, then using drag gestures on the screen to position the flower into their preferred position in the garden. This flower, with its corresponding note, location, and flower type, is persistent as it is saved permanently in the garden. Anyone else who visits the garden will now see that flower. Dream Garden was implemented using SwiftUI¹ and ARKit² as the front-end, and Firebase³ for the database. Flower assets come from Sketchfab⁴ in the form of USDZ files.

To describe our system through a vignette, Lisa opens Dream Garden on her phone while walking by a park. She places a flower there with the note "sustainability for Earth." A few days later, Ian, who doesn't know Lisa, passes by the same place and opens the app. He sees Ian's flower, taps on it, and reads Lisa's note. He leaves his own flower. Over the following weeks, the park gets populated with hundreds of flowers and notes from passers-by, and Lisa and Ian may return and see how the garden has changed since their previous visit.

4 EVALUATION

We recruited ten university students from different majors and years, sourced from personal outreach to students by the authors, to visit the Dream Garden. We asked participants to come to a specific location, we chose a big open field at our university campus. We handed a phone with the app preinstalled, asked them to open the app, and then explore the digital garden by tapping on flowers. We planted the initial flower, so the first participant saw one flower. We interviewed them using a semi-structured interview protocol (see appendix) that included questions such as "How does seeing these flowers make you feel?" and "Do you feel connected to these other people? In what ways?". We then asked participants to plant their own flowers, including a note corresponding to a dream they have for "themselves or the world." We then interviewed them about how they feel after planting the flower. Their flower will now be visible to all future participants.

We asked the first five participants to return to the garden one week later (after all ten had their first session). They saw eleven flowers, in contrast to the 1-5 flowers they saw in session one. We asked them to explore the garden again and voice their experience,

conducting a semi-structured interview as shown in the appendix. The participants never interacted with the garden at the same time.

5 STUDY RESULTS

We found that participants were excited about building a space with strangers, liked the dynamic nature of the garden, felt connected to the garden location, and felt a sense of connection to others, though in different ways.

5.1 Participants felt a sense of joy and belonging from participating in building a shared space, but not necessarily a sense of emotional connection.

Participants would run around the flowers, circling them and getting in for a closer look. Most were smiling when they were exploring, making comments about how fun the experience was, and contributing to the experience by planting their own flowers. As P5 put it, *"I feel like I am playing a video game with strangers... seeing the flowers makes me feel happy."* P7 put it as, *"It feels more like I'm part of a community... we are all doing something together."* However, though participants felt a sense of belonging, many did not particularly feel connected to the other participants as the flowers were anonymous and ranged in depth. For example, P4 mentioned, *"I don't have enough of a sense of what people are doing... I feel the most connection to the most substantial messages."* Participants felt more connected to longer, more thoughtful notes left by strangers, rather than short notes such as "joy." The more thoughtfulness a participant put into their flower, the more connected others felt to them, and feelings of connectedness varied among participants. Some participants took a few minutes with each flower, reflecting deeply on the associated text. Others sped through the flowers and were more excited about planting their own.

5.2 Garden location became meaningful to participants.

The location of the garden became meaningful to the participants, and they felt bonded to the location being the main University field. As P2 put it, *"This location is the symbol of the hopes and dreams of students... maybe with the flowers we will create a strong dream energy that can elevate all of our dreams."* Others commented about wanting to return to the garden location in their own time to reflect and see how the garden changes.

5.3 Participants expressed a desire to connect with strangers, but were frustrated with the lack of ways to do so.

Often participants would tap a flower and read the dream, then pause to think about the dream. Some wondered out loud what a certain dream meant. For example, one person wrote 'ocean waves,' and many wondered what they meant. Many expressed that they hoped their flowers would relate to the participants that came after them. Others expressed a desire to react to flowers in some way, such as watering them and caring for their dreams. P6 said, *"I want to be able to give other people's dreams some care... to water their flowers."* Others wanted to reply to dreams, adding their own

¹SwiftUI is an Apple framework for creating mobile apps on iOS.

²ARKit is an Apple framework for creating augmented reality experiences for iOS.

³Firebase is a Google-made database that has a convenient interface, we use it to store each flower information and data persistence.

⁴Sketchfab is an online platform that allows you to publish, share, and discover 3D assets online.

thoughts to flowers. Our system did not have a way for people to respond to others' flowers, and participants expressed frustration at the lack of ways to continue the discussion. Some responded by making their flowers similarly themed to other flowers, and placing them next to their favorite flowers.

5.4 Participants were excited about the experience changing over time, and expressed a desire to return to the garden.

P6 said, *"I'd like to be able to go back to it on my own time and watch it grow, and see what people change as they plant their flowers."* While usual AR experiences are static and do not change over time, this sense that participants would return to the same experience is very significant. The participants who returned for a second session were particularly excited, as they only saw a few flowers in the initial session. Participants made comments such as *"I want to see more of them!"* (P1) or compared the flowers to *"footprints"* (P2).

5.5 Physical embodiment was conducive to participants feeling invested in the experience.

The physical embodiment of the garden - even if it was virtual - was expressed by multiple participants. Many commented that if this was just on a website, they would not be as emotionally connected to the garden or the other participants: *"I feel more connected to a broader community than I feel online... it feels more real and communal because we were all standing in this spot... we all share this commonality in space instead of the ether of Twitter"* (P2) Others reached for the flower as though it was there in reality. Having a flower and its dream metaphorically take up space made it more significant for participants, increasing the sense of connection they felt to the experience. This was portrayed as unique from web alternatives, such as Twitter posts, which do not have this sense of physical embodiment.

6 LIMITATIONS

We identified several opportunities for building location-based, participatory AR systems; however, our findings come from a small and homogeneous group of participants, college students, who are heavy users of technology and already have a shared context of being classmates. Future work should explore recruiting a more diverse group of participants. Further, our system only allowed participants to interact with the garden in specific ways, and exploring new interaction methods, such as responding to flowers or watering them, would be beneficial. In addition, the decision to make the location-based participatory space a garden with the theme of dreams was driven by finding a theme everyone can understand, and exploring other themes and comparing the results would further expand this research.

One of our main limitations is that our interview process did not include directly asking participants about their social well-being prior to the AR experience. This limits our understanding of the change of feelings of connection that our system encourages and our assessment of feelings of connection. In the future, a before

and after study of feelings of social well-being, using an established questionnaire, would further this research.

7 DISCUSSION

We found that location-based experiences help people associate a certain meaning to a place, giving the digital experience more significance. AR gives the experience a feeling of embodiment, with participants feeling more connected to the garden due to the garden taking up physical space in contrast to potential non-AR alternatives. We also found that spaces people build together elicit joy in participants, and participants were excited about the space changing over time and being part of that change.

Regarding participants feeling connected to each other, we found a tension between creating a place for reflection and one for interaction. Participants reflected on the *"snippet of someone else"* they saw in each flower, and for some, this was a very reflective experience. Those participants felt connected to others that were more thoughtful in the creation of their flower. Others were more excited by the prospect of growing a garden together and wanted to reply to others and interact with their flowers. Each of these brings up a different sense of connection, with emotional connection being important to the former group, and belonging to the latter group.

We think this tension between emotional connection and joyful belonging stems from the theme of the space we chose, which is more reflective in nature, and our lack of a specific definition of what type of space the garden is. This brings up future research opportunities for designing these types of AR experiences.

8 CONCLUSION AND FUTURE WORK

This research explores location-based, participatory experiences that help strangers connect to each other and, eventually, contribute to improving the social well-being of our society. Our study shows promising results toward these goals.

In our study, we learned that participants established a sense of connection to the physical location of the garden and that the use of AR facilitated being more invested in the experience as a whole, reproducing previous research on place attachment. Furthermore, we found that people were excited about participating in the co-creation of the garden and expressed a desire to meet other participants. This suggests that these digital experiences may act as catalysts for in-person interaction rather than replacements for in-person interaction. Through co-creation and sharing personal information (their dreams), we found that participants felt a sense of connection to the other participants. This connection is one of the most exciting outcomes of this research, as it offers an opportunity to explore ways of bringing people together through AR, a medium often associated with individuality and distraction.

Although AR is still in its early stages, it offers exciting opportunities and directions for research. Some specific aspects that we think should be explored further, inspired by this research, are how AR offers the possibility of creating meaningful connections to a certain location and how AR may act as a mediator for initiating interaction between community members. For example, looking into dynamic, ever-changing spaces as a way to allow strangers to connect, incorporating the idea of seeing snippets of a stranger's

personality as a way to increase a sense of community, and exploring the possibilities of collaboratively created persistent digital spaces and how they may contribute to a sense of joy and belonging among strangers would be exciting directions for future research.

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A INTERVIEW PROTOCOL

Below is the process we asked participants to follow during the study:

- (1) Each of the 10 participants was asked to meet at the university field.
- (2) The participant was asked to explore flowers in the garden. Then they were asked the following questions:
 - What are your initial thoughts on the garden?
 - How does seeing these flowers make you feel?
 - How does seeing these notes make you feel?
 - Do you feel connected to these other people? In what ways?
- (3) The participant was then asked to plant their own flower, then asked the following questions:
 - How do you feel after planting the flower?
 - Other people will see your flower and note, how do you feel about this?
 - Do you feel differently towards the other flowers and people who planted them?
 - What are your thoughts on the dream garden as a whole?
- (4) The first 5 participants were asked to return to the field a week later for a second session. In this session they were asked to explore the now-grown garden and asked the following questions:
 - Do you feel different coming back to the garden?
 - What changes do you notice in the garden?
 - How do you feel about the new flowers that were planted since your last visit?

- Do you notice yourself interacting differently with the garden?